

Contact

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Top Skills

System Architecture
Business Process Automation
Cloud Services

Certifications

Introduction to HTML, CSS, & JavaScript
Introduction to Software Engineering
Supervised Machine Learning: Regression and Classification
Programming with JavaScript
Introduction to Front-End Development

Alexey Martinez

Full-Stack Systems Engineer | Building Practical Software, AI Workflows & Internal Tools for Business Operations
Ann Arbor, Michigan, United States

Summary

I'm a bilingual (English/Spanish) software and systems engineer working across full-stack development, operations, AI integration, and industrial automation.

Over the past 5 years, I've gone from building internal tools solo to leading teams delivering enterprise-scale systems. I'm currently leading a team building a full ERP system while also owning a separate operations platform I built from scratch that serves the entire company daily.

What I do well: I look at any process, whether it's an industrial workflow or a business operation, identify what's broken or inefficient, and build something that fixes it. That's looked like an AI-powered receipt analysis module that automates expense matching, a data pipeline that communicated directly with FANUC industrial robots to push process reliability from 85% to 99%.

I work across the full stack, Node.js, React, ASP.NET, Java, Angular, Python, Docker, and Linux infrastructure. I've also adopted AI-assisted, spec-driven development using Claude and OpenAI Codex, which lets me design, architect, and ship production systems significantly faster without cutting corners on quality.

Always open to connecting with builders, founders, and teams doing interesting work in tech, automation, and AI.

Experience

Encore Automation

4 years 10 months

Systems Engineer

June 2025 - Present (11 months)

Auburn Hills, Michigan, United States

Leading a team of 2 engineers designing and building a full enterprise ERP system (ASP.NET, Angular).

Individually architected and delivered an internal operations platform from scratch that serves all company employees daily, replacing manual spreadsheet workflows with automated timesheet entry, expense management, event scheduling, deadline tracking, and multi-level approval routing.

Integrated an AI receipt analysis module using OpenAI Vision that extracts structured line-item data from uploaded receipts and auto-matches entries, reducing manual review time for administrators.

Built a Python application that interfaced directly with FANUC industrial robots to extract real-time pressure sensor data and flag faulty sealed fasteners, improving process reliability from ~80% to 99%.

Own all production infrastructure on self-hosted Ubuntu servers, Docker deployments, CI/CD pipelines, reverse proxy configuration, maintaining 99%+ uptime across all internal platforms.

Product Development Engineer

July 2021 - June 2025 (4 years)

Michigan, United States

Designed and developed a patent-pending robotic end-of-arm tool that improved the speed and precision of sealant application on moving automotive lines, now deployed across multiple plants in the U.S. and Canada.

Led iterative prototyping and development of multiple robotic tools, directly expanding the company's product lineup and reducing time-to-deployment for new industrial solutions.

Reverse-engineered and optimized mechanical components to reduce material and manufacturing costs while improving adaptability for custom industrial applications.

Owned full-cycle system installations at customer plants working directly with plant managers and operators to ensure reliable integration, and building the kind of on-site relationships that brought repeat business.

During this role, began transitioning into software development building internal tooling, automation scripts, and data systems that eventually became the foundation for the company's internal software platform.

GE Appliances, a Haier company

Technology Co-Op

January 2020 - June 2020 (6 months)

Louisville, Kentucky, United States

Worked on AC Ductless Systems, testing & designing product components. Assisted in product development & efficiency testing for HVAC solutions.

Florida International University

Statics Learning Assistant

January 2019 - December 2019 (1 year)

Taught students to apply Calculus and Physics concepts to solve complex engineering problems.

Collaborated with the professor to guide students in class, enhancing their understanding of engineering fundamentals.

Conducted tutoring sessions to assist students and answer course-related questions.

Education

Florida International University - College of Engineering & Computing

Bachelor of Science - BS, Mechanical Engineering (Robotics

Focus) · (2018 - May 2021)